

Monash University, Faculty of Information Technology

FIT3080

Applied Class: Bayesian Networks Solutions

Please note: Use a calculator or spreadsheet where helpful.

Exercise 1: Joint probabilities, conditional probabilities and Bayes' theorem

When driving into the city, you've noticed a pattern that there is often heavy traffic when there is a major event or it is peak hour. You estimate the joint probabilities as:

MajorEvent (M)	PeakHour (H)	Traffic (T)	P()
Yes	Yes	Heavy	0.005
Yes	Yes	Moderate	0.001
Yes	Yes	Light	0.000
Yes	No	Heavy	0.010
Yes	No	Moderate	0.020
Yes	No	Light	0.004
No	Yes	Heavy	0.085
No	Yes	Moderate	0.045
No	Yes	Light	0.014
No	No	Heavy	0.080
No	No	Moderate	0.240
No	No	Light	0.496

(All answers to 3 decimal places.)

- What is the probability that traffic will be heavy $P(T=\text{Heavy})$?
 Answer: $0.005+0.010+0.085+0.080=0.180$
- What is the probability that traffic will be moderate or heavy $P(T=\text{Heavy} \vee T=\text{Moderate})$?
 Answer: 0.180 (i.e. from before) $+ 0.001+0.020+0.045+0.240=0.486$
- What is the probability that there is a major event and it is peak hour $P(M=\text{Yes} \wedge H=\text{Yes})$?
 Answer: $0.005+0.001+0.000 = 0.006$
- What is the probability that there is a major event if traffic is heavy $P(M=\text{Yes}|T=\text{Heavy})$?
 Answer: $(0.005+0.010)/0.180 = 0.083$

- e) Is the level of traffic probabilistically dependent on a major event?

We can test for a probabilistic dependence by checking if $P(T|M) \neq P(T)$ or if $P(M|T) \neq P(M)$.

If we check $P(T|M) \neq P(T)$, e.g., when $T=Heavy$ and $M=Yes$:

$$P(T=Heavy|M=Yes) = (0.005+0.010)/(0.005+0.001+0.010+0.020+0.004) \\ = 0.015/0.04 = 0.375$$

$$P(T=Heavy) = 0.005+0.010+0.085+0.080=0.18$$

Different in at least one case means they are dependent

Or we can check $P(M|T) \neq P(M)$. e.g., again when $T=Heavy$ and $M=Yes$:

$$P(M=Yes)=0.005+0.001+0.000+0.010+0.020+0.004=0.04, \text{ but}$$

$$P(M=Yes|T=Heavy)=0.083 \text{ (from d)}$$

So again, difference in at least one case leads us to conclude there is dependence.

- f) Show the conditional distribution for major events and traffic, given it is peak hour $P(M \wedge T | H=Yes)$.

MajorEvent (M)	Traffic (T)	P()	Normalised P()
Yes	Heavy	0.005	0.0333
Yes	Moderate	0.001	0.0067
Yes	Light	0.000	0.000
No	Heavy	0.085	0.5667
No	Moderate	0.045	0.3000
No	Light	0.014	0.0933
TOTAL		0.15	

- g) (Optional) Are major events dependent on whether it is peak hour?

Answer: No. We can show this by showing that 1) all the conditioning cases are equal, i.e. $P(M|H=Yes) = P(M|H=No)$, or 2) that $P(M|H)=P(M)$.

For approach (1), we can start with:

$$P(M=Yes|H=Yes) =$$

$$(0.005+0.001+0.000)/(0.005+0.001+0.000+0.085+0.045+0.014) = 0.04$$

&

$$P(M=Yes|H=No) =$$

$$(0.010+0.020+0.004)/(0.010+0.020+0.004+0.080+0.240+0.496) = 0.04$$

then, for $M=Yes$, the two conditioning cases produce the same probability.

Since M is binary, then $P(M=No|H) = 1 - P(M=Yes|H)$, which will be $1 - 0.04 = 0.96$ for both $H=Yes$ and $H=No$ as well. Hence there is no dependence.

For approach (2), we can calculate:

$$P(M=Yes | H=Yes) =$$

$$(0.005+0.001+0.000)/(0.005+0.001+0.000+0.085+0.045+0.014) = 0.006/0.15 = 0.04$$

$$P(M=Yes) = 0.005+0.001+0.000+0.010+0.020+0.004 = 0.040$$

So $P(M=Yes|H=Yes) = P(M=Yes)$. Since M is binary, then $P(M=No|H=Yes) = 1 - P(M=Yes|H=Yes) = 0.96$ and $P(M=No) = 1 - P(M=Yes) = 0.96$. Since H is binary, and P(M) is some weighted average of $P(M|H)$, i.e., there is an $\alpha = P(H=Yes)$ such that:

$$P(M) = \alpha P(M|H=Yes) + (1-\alpha)P(M|H=No)$$

Since we calculated that $P(M) = P(M|H=Yes)$, then:

$$P(M) = \alpha P(M) + (1-\alpha)P(M|H=No)$$

$$(1-\alpha)P(M) = (1-\alpha)P(M|H=No)$$

$$P(M) = P(M|H=No)$$

which shows that both sides are equal. (Or we could calculate it directly from the table to check.)

Exercise 2: Conditional probabilities and Bayes' theorem (Medical condition)

Approximately 1 in 20 people in Australia have type 2 diabetes. An oral glucose tolerance test (OGTT) is a way of testing for the presence of type 2 diabetes and is considered quite accurate. Numerical estimates of its accuracy vary, but are generally close to a true positive rate of 85% and a false positive rate of 5%.

Apply Bayes' theorem to answer the following questions:

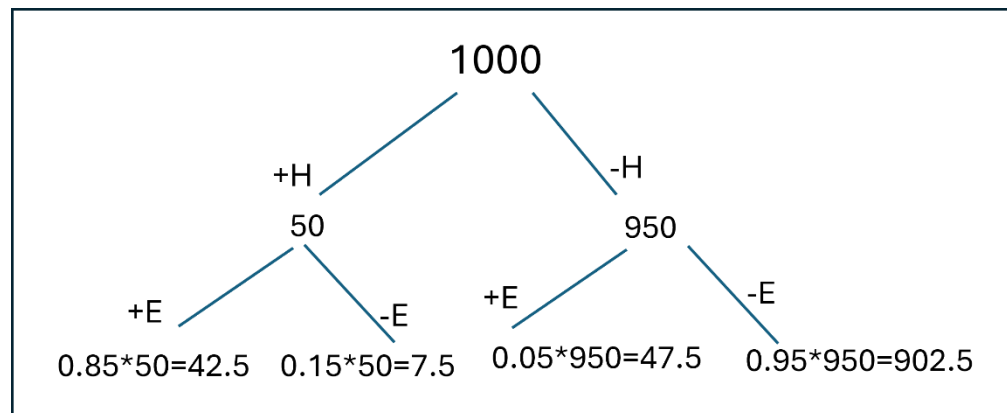
- a) If someone tests positive on the OGTT then what is the probability that they have diabetes?

Answer: If H is "has type 2 diabetes", E is "tests positive on OGTT", then:

$$P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H) = 0.85 \cdot 0.05 + 0.05 \cdot 0.95 = 0.09$$

$$P(H|E) = P(E|H)P(H)/P(E) = 0.85 \cdot 0.05 / 0.09 = 0.472 = 47\%$$

Using natural frequencies:



$$P(+H | +E) = 42.5 / (42.5 + 47.5) = 42.5 / 90 = 0.472$$

- b) If someone tests negative on the OGTT then what is the probability that they have diabetes?

Answer:

$$P(\neg E) = P(\neg E|H)P(H) + P(\neg E|\neg H)P(\neg H) = 0.15 \cdot 0.05 + 0.95 \cdot 0.95 = 0.91$$

$$P(H|\neg E) = P(\neg E|H)P(H)/P(\neg E) = 0.15 \cdot 0.05 / 0.91 = 0.008 \approx 1\%$$

Using natural frequencies:

$$P(+H | -E) = 7.5 / (7.5 + 902.5) = 7.5 / 910 = 0.008$$

- c) Given the results, should we rely on the OGTT alone in diagnosing diabetes? In more realistic situations, why might our prior be different to the general prevalence of diabetes?

Answer:

<1% is the posterior $P(H|\neg E)$ and approx 50% is the posterior $P(H|E)$. 1% is

quite certain, 50% isn't. So, it may be reasonable to rely on OGTT only for negative results, and not for positive results.

The general prevalence of diabetes is how common diabetes is in the population (1/20). Someone who goes/is sent to have an OGTT to test if they have diabetes is *not* representative of the general population, so the prior is probably higher than 1/20 (due to selection bias). If someone is having a test, there may be other signs and symptoms that suggested a test would be good to do in the first place. Those other signs and symptoms would also increase our probability estimate.

- d) How would a false positive rate of 1% change our estimate of the probability that someone has diabetes if they test positive?

Answer:

$$P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H) = 0.85 \cdot 0.05 + 0.01 \cdot 0.95 = 0.052$$

$$P(H|E) = P(E|H)P(H)/P(E) = 0.85 \cdot 0.05 / 0.052 = 0.817 = 82\%$$

It would increase the probability of having diabetes from 47% to 82%. This would significantly change the decisions we might make.

- e) (Optional) Is the test more sensitive to changes in the true positive rate or the false positive rate?

Answer:

In (d), we tried reducing the false positive rate (FPR) to 0.01. Let's reset the FPR to 0.05 and now try reducing the TPR by the same amount (-0.04) to 0.81:

$$P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H) = 0.81 \cdot 0.05 + 0.05 \cdot 0.95 = 0.088$$

$$P(H|E) = P(E|H)P(H)/P(E) = 0.81 \cdot 0.05 / 0.088 = 0.460 = 46\%$$

The probability of having diabetes went from 47.2% to 46%, so the test is **much** more sensitive to (additive) changes in the false positive rate.

Exercise 3: Bayesian Network (Late to Campus)

Let's assume that you don't have a car, but rather you come to campus either by bike or by bus. If you take the bus, there is a 20% chance that you will be late. If you take the bike, there is a 5% chance that you will be late. You come to campus by bike 4 days out of 5. Today you were late to campus.

- a) Write down and explain the formula used in Bayes' rule for this problem:

$P(\text{bus}|\text{late})$

Answer:

$$P(\text{bus}|\text{late}) = P(\text{late}|\text{bus})P(\text{bus})/P(\text{late}).$$

That is, we update the prior for taking the bus $P(\text{bus})$ by multiplying by the likelihood $P(\text{late}|\text{bus})$ to obtain the joint $P(\text{bus}\&\text{late})$, and divide by $P(\text{late})$, to condition on late (i.e. $\text{late}=\text{T}$ and $\text{late}=\text{F}$).

- b) Use Bayes' rule to calculate the probability that you took the bus today.

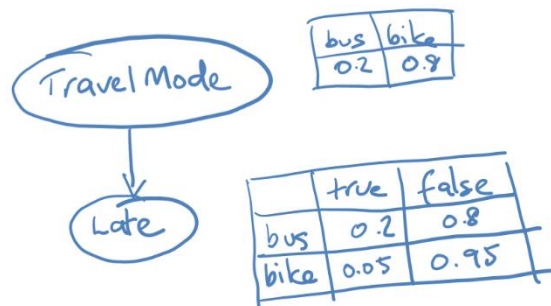
Answer:

$$P(\text{late}) = P(\text{late}|\text{bus})P(\text{bus}) + P(\text{late}|\neg\text{bus})P(\neg\text{bus}) = 0.2 \cdot 0.2 + 0.05 \cdot 0.8 = 0.08$$

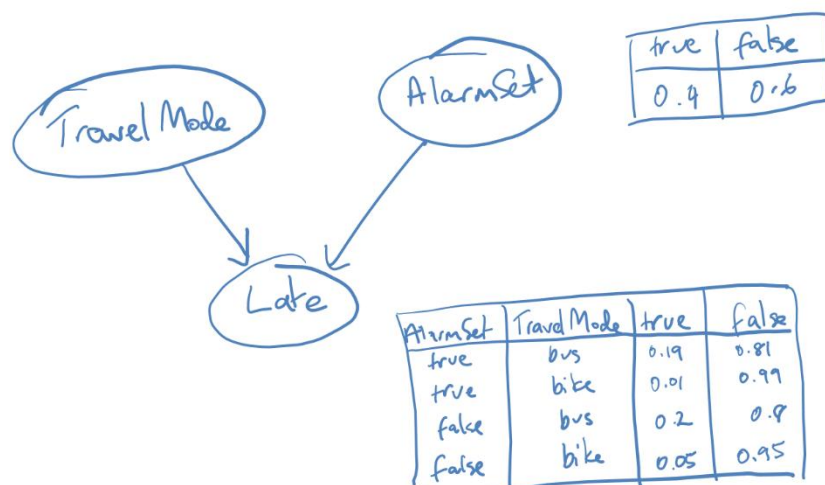
$$P(\text{bus}|\text{late}) = P(\text{late}|\text{bus})P(\text{bus})/P(\text{late}) = 0.2 \cdot 0.2 / 0.08 = 0.5$$

- c) Model the situation as a Bayesian network with 2 nodes and give the conditional probability distribution for both nodes.

Answer:



- d) Add a third node for setting an alarm. You set an alarm 40% of the time, and it reduces your chances of being late to 1% if travelling by bike and 19% if travelling by bus.



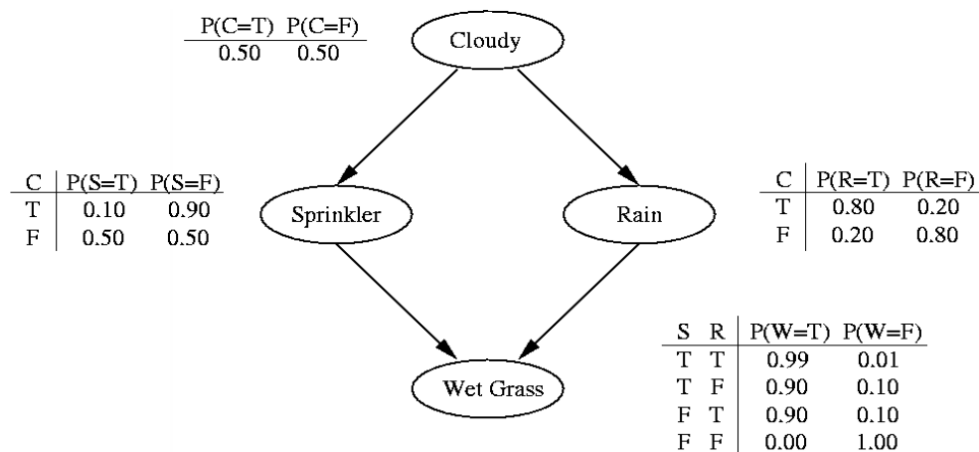
- b. Without calculating, will $P(\text{bus}|\text{late}, \text{alarm}=\text{True})$ be higher, lower or the same as $P(\text{bus}|\text{late}, \text{alarm}=\text{False})$?

Answer:

Higher. It's now even more likely that you *will be* late if you came by bus than by bike, hence if you *are* late, you are now even more likely to have come by bus.

Exercise 4: Bayesian Network (Sprinkler)

Consider the following Bayesian network:



The event that the grass is wet ($W=\text{true}$) has 2 possible causes. It is either the water sprinkler was on ($S=\text{true}$) or it rained ($R=\text{true}$).

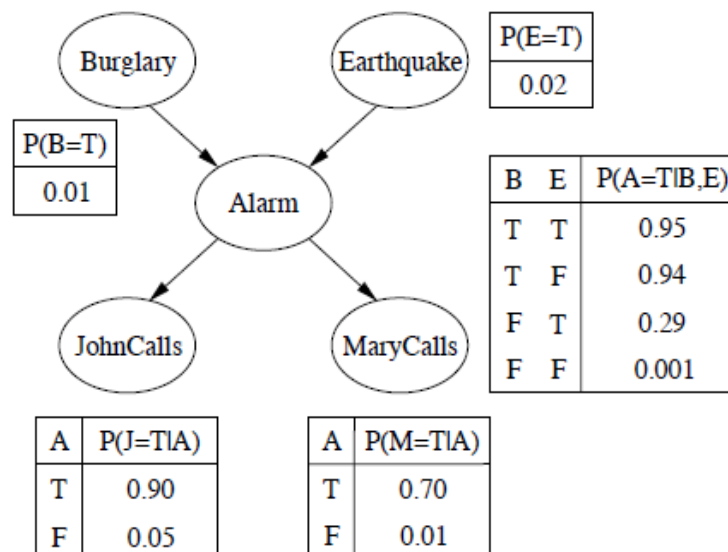
Construct this BN using Netica software. Use it to help you answer the following queries:

- Given the observation that the grass is wet, what is the probability that it is caused by rain?
Answer: Enter evidence for Wet Grass, and check the probability of Rain=T, giving 0.708
- Given the observation that the grass is wet, how likely is it due to the water sprinkler was on?
Answer: Leaving the same evidence entered, check the probability of Sprinkler=T, giving 0.43

Exercise 5: Bayesian Network (Pearl's Earthquake example) (Netica)

Example statement: You have a new burglar alarm installed. It reliably detects burglary, but also responds to minor earthquakes. Two neighbors, John and Mary, promise to call the police when they hear the alarm. John always calls when he hears the alarm, but sometimes confuses the alarm with the phone ringing and calls then also. On the other hand, Mary likes loud music and sometimes doesn't hear the alarm. Given evidence about who has and hasn't called, you'd like to estimate the probability of a burglary (from Pearl (1988)).

A BN representation of this example is shown in Figure 2.6. All the nodes in this BN are Boolean, representing the true/false alternatives for the corresponding propositions. This BN models the assumptions that John and Mary do not perceive a burglary directly and they do not feel minor earthquakes. There is no explicit representation of loud music preventing Mary from hearing the alarm, nor of John's confusion of alarms and telephones; this information is summarized in the probabilities in the arcs from *Alarm* to *JohnCalls* and *MaryCalls*.



[Above taken from Korb&Nicholson, Bayesian Artificial Intelligence, Section 2.5.1]

Construct an Earthquake BN using Netica software. Use it to help you answer the following questions about specific scenarios. You can also consult slides 70-73 from Week 6 for conditional dependence definitions.

- Only Earthquake is set to True: which nodes are not affected and why?
 Answer: Only Burglary is not affected. Without knowing about the Alarm, Earthquake is independent from Burglary.
- Only Alarm is set to True: which nodes are not affected and why?
 Answer: all nodes are affected.

(c) With Alarm set to True, Earthquake is set to False:

(i) which nodes are affected and why?

Answer: Only Burglary is affected: Given that the Alarm has rung, there being no Earthquake indicates that there must have been a Burglary.

(ii) what is the relationship between Earthquake and Burglary?

Answer: They are both causes of Alarm.

(iii) which nodes are not affected and why?

Answer: The effect of Earthquake on the downstream nodes is blocked by the Alarm.

(d) Answer these three questions with Alarm set to False.

(i) which nodes are affected and why?

Answer: With the Alarm set to False, both Earthquake and Burglary are extremely unlikely. Setting Earthquake to False doesn't change that fact about Burglary.

(ii) what is the relationship between Earthquake and Burglary?

Answer: They are both causes of Alarm. If neither occurred, there is no Alarm.

(iii) which nodes are not affected and why?

Answer: The effect of Earthquake on the downstream nodes is blocked by the Alarm.

(e) With Alarm set to True, John Calls is set to False: which nodes are affected and why?

Answer: No nodes are affected, as Alarm blocks the propagation of evidence from JohnCalls.